INDIVIDUAL PROJECT / DATA ANALYSIS

Exploratory Data Analysis

Indian Car Market Dataset

A clear look at 10,000 car records, with attention to data quality, prices, vehicle features, and the limits of the dataset.

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Dataset source	Kaggle - Indian Car Market Dataset

Project focus
Explore the data as it is, report what it can support, and avoid treating a pattern as proof of cause.

The assignment does not require a machine-learning model. This report focuses on exploratory analysis and evidence-based observations.

1. Objective and dataset

The purpose of this project is to understand the records in the car dataset before drawing conclusions from them. The analysis checks completeness, summarizes the main fields, and compares price with selected vehicle characteristics.

Questions used in this report

- How are listed prices distributed?
- Which brands and fuel types appear most often?
- How do listed prices compare by transmission and brand?
- Do mileage or model year show a clear relationship with price?

Dataset overview

The CSV contains **10,000 rows and 11 columns**: an identifier, four categorical fields, and six numeric measurements. The Kaggle link is the source recorded in the supplied Python file.

Field	Type	Description from the field name
Car_ID	Integer	Record identifier; not a vehicle measurement.
Brand	Category	Manufacturer label.
Model	Category	Car model name.
Year	Integer	Model year.
Fuel_Type	Category	Recorded fuel category.
Transmission	Category	Manual or automatic label.
Price	Decimal	Price value; currency is not stated.
Mileage	Decimal	Mileage value; unit is not stated.
Engine_CC	Integer	Engine size in cc, as indicated by the column name.
Seating_Capacity	Integer	Number of seats.
Service_Cost	Decimal	Service cost value; unit is not stated.

Sample records

ID	Brand	Model	Year	Fuel	Transmission	Price
1	Toyota	Innova	2024	CNG	Manual	2,020,000
2	Kia	EV6	2023	Diesel	Manual	1,770,000
3	Maruti Suzuki	Dzire	2016	Petrol	Manual	3,430,000

The sample is shown as stored in the CSV. Descriptions and units are limited to what the file provides.

2. Data quality and cleaning

Cleaning means checking for problems and correcting them only when the data supports a clear decision. The basic completeness checks pass, so no rows need to be filled or removed.

Check	Result	Decision
Rows and columns	10,000 rows x 11 columns	Starting dataset size recorded.
Missing values	0 in every column	No imputation needed.
Exact duplicate rows	0	No duplicate rows removed.
Category spelling/spacing	No leading/trailing spaces or case variants in the four categories.	No label normalization needed.
Brand and model pairing	Each of 50 models maps to one brand in this file.	No brand/model changes made.
Model and fuel pairing	Each model occurs with all 4 fuel labels.	Flagged as a concern; no trusted mapping was available.
Numeric data types	Year, Price, Mileage, Engine_CC, Seating_Capacity and Service_Cost are numeric.	No type correction needed.
Possible IQR outliers	0 flagged across the six numeric measurements.	No values removed; this is a screening check.
Rows after checks	10,000	No row count change: 10,000 before and after.

Why the fuel pairing is flagged

There are 50 model names, and each appears with every fuel label: CNG, Diesel, Electric, Petrol. For example, the sample includes a Kia EV6 with Diesel. This is a cross-field warning, not proof that every row is wrong. Without a trusted data dictionary or source mapping, changing the records would be guesswork.

Outlier rule: a value is flagged below Q1 – 1.5 × IQR or above Q3 + 1.5 × IQR. Car_ID is excluded because it identifies rows rather than measuring a car feature.

Cleaning outcome: 10,000 rows before and after. No missing values were filled, no duplicate rows were dropped, and no potential outliers were removed. The fuel pairing concern is documented for interpretation.

3. Descriptive analysis

The model years cover 2015–2024. Price values range from 400,000 to 3,500,000 in the units recorded by the CSV. The mean and median are close, while brand and fuel counts are fairly evenly spread.

Numeric summary

Measure	Mean	Median	Minimum	Maximum
Year	2,019.54	2,020.00	2,015	2,024
Price	1,946,064	1,950,000	400,000	3,500,000
Mileage	19.97	20.00	10.00	30.00
Engine_CC	1,542.07	1,500	800	2,500
Seating_Capacity	5.52	6.00	4	7
Service_Cost	14,969	15,000	5,000	25,000

Categorical summary

Field	Distinct values	Most common	Least common
Brand	10	Maruti Suzuki (1,042)	Toyota (954)
Model	50	Amaze (239)	WR-V (163)
Fuel_Type	4	CNG (2,588)	Petrol (2,426)
Transmission	2	Manual (5,045)	Automatic (4,955)

The average price is 3,936 units below the median. Brand counts range from 954 to 1,042; fuel counts range from 2,426 to 2,588. The model/fuel pattern means the fuel mix should not be treated as a verified picture of real vehicle specifications.

Limitations

The CSV does not state the currency, mileage unit, collection method, region, or listing date. The model/fuel combinations need external validation. These limits narrow what can be said about the wider car market; this report describes the file as supplied.

The report keeps recorded numeric values and avoids assigning undocumented units.

4. Visual EDA: price and dataset mix

Each chart has a short observation. These notes describe this dataset; they do not imply a cause.

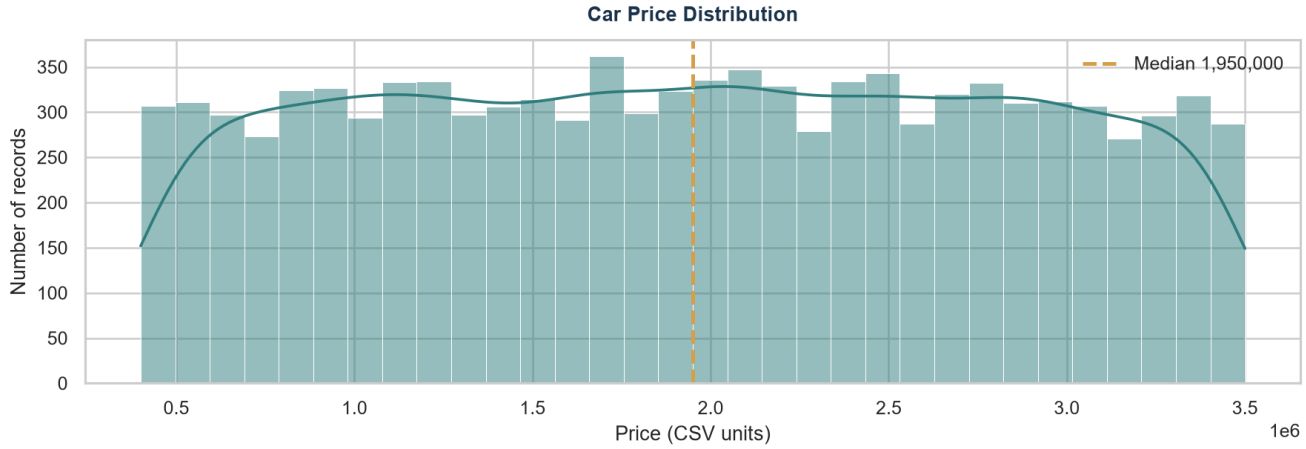


Chart 1. Price distribution. The mean (1,946,064) is close to the median (1,950,000); values span 400,000–3,500,000 in the CSV units.

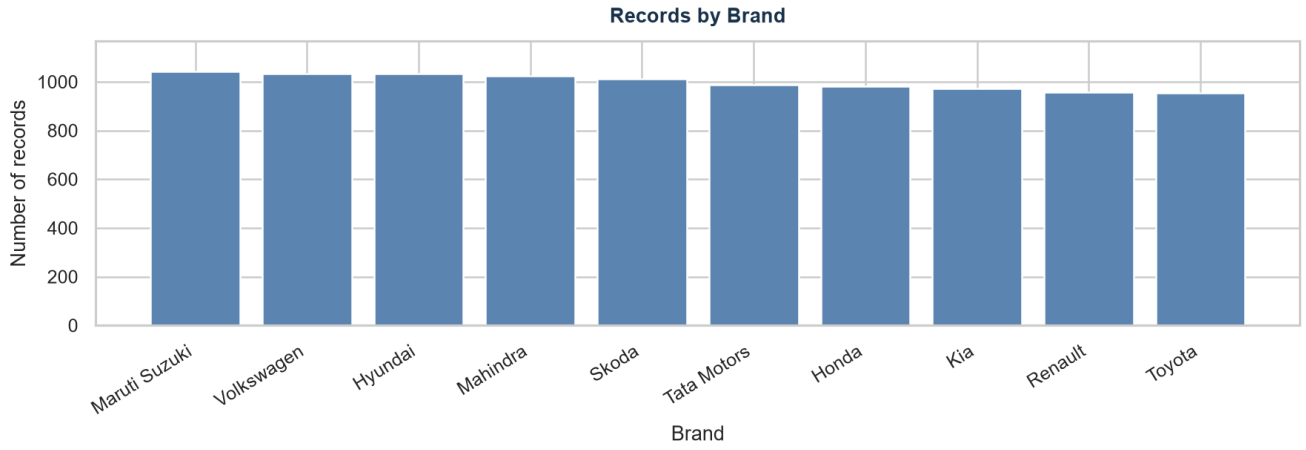


Chart 2. Brand counts. Maruti Suzuki is most represented (1,042); Toyota is least represented (954).

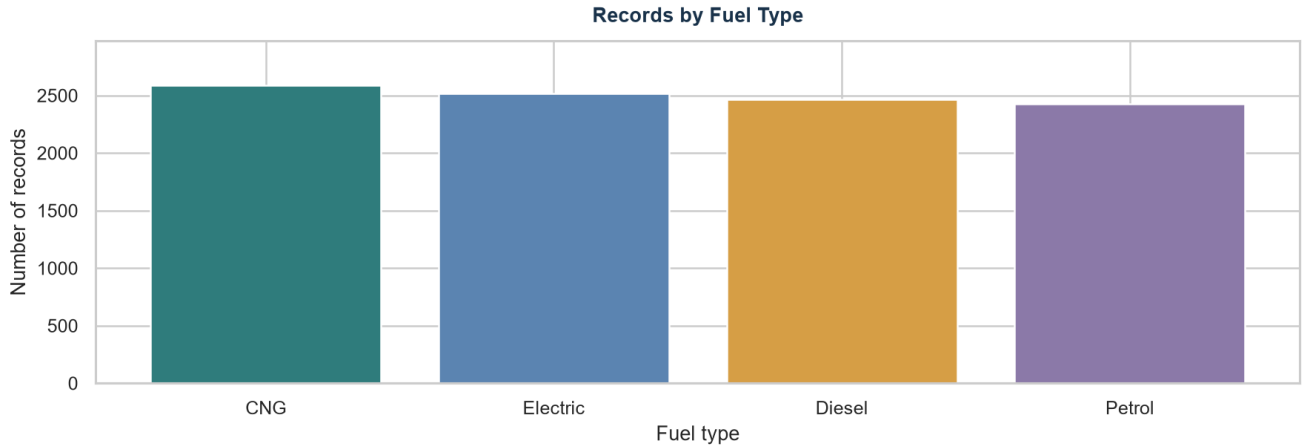


Chart 3. Fuel-type counts. CNG is most common (2,588) and Petrol least common (2,426). Read this alongside the model/fuel concern on page 3.

4. Visual EDA: price relationships

The comparisons use values as recorded. Correlation describes a linear association and does not establish causation.

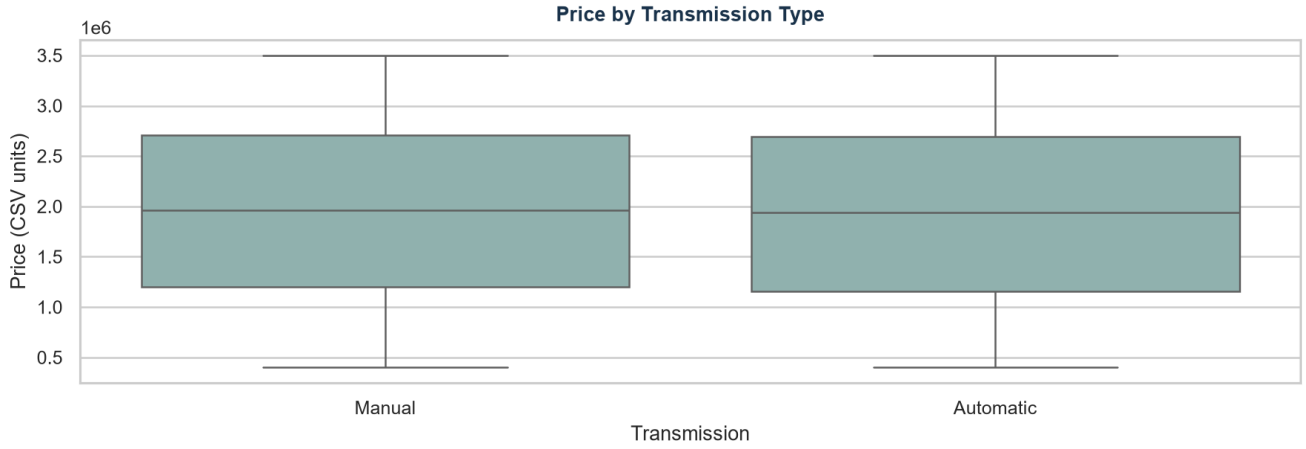


Chart 4. Price by transmission. Median price is 1,960,000 for Manual and 1,940,000 for Automatic cars, a small descriptive difference.

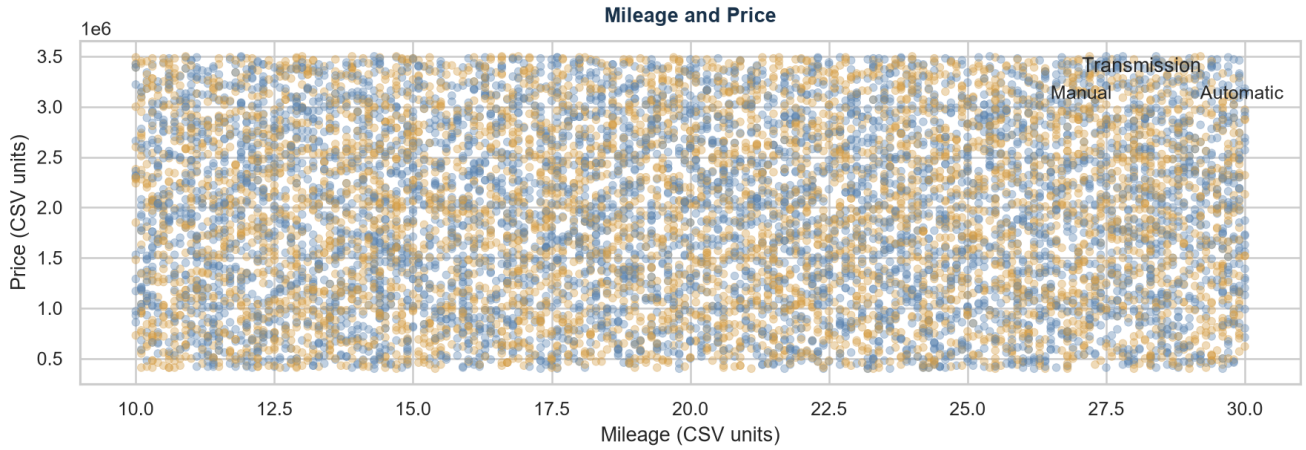


Chart 5. Mileage and price. Pearson r = 0.013, close to zero; the points show no clear linear trend.



Chart 6. Model year and price. Pearson r = -0.002, also close to zero in this file.

4. Visual EDA: brand prices and numeric relationships

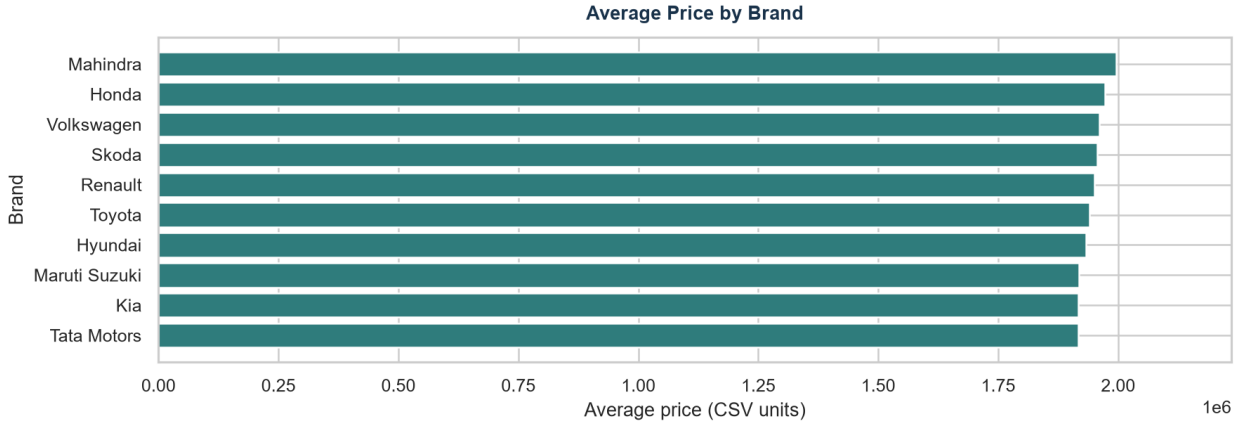


Chart 7. Average price by brand. Mahindra is highest (1,996,396); Tata Motors is lowest (1,916,198). This is descriptive.

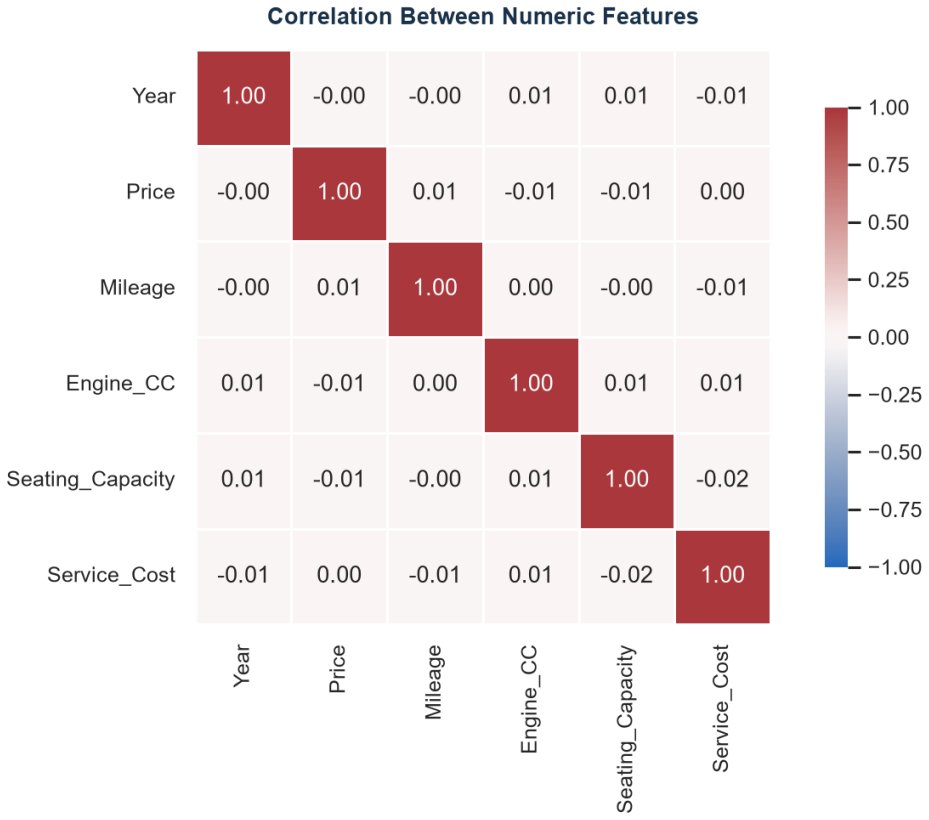


Chart 8. Correlation heatmap. The largest absolute off-diagonal value is -0.016 (Seating_Capacity and Service_Cost), so the numeric fields show little linear association.

Key findings

1. The 10,000-row file has no missing values and no exact duplicate rows.
2. Brand representation is close across 10 labels, ranging from 954 to 1,042 records.
3. Fuel labels are similarly balanced, but every model appears with all fuel types; this weakens real-world interpretation of the fuel totals.
4. Price mean and median are close; the recorded range is 400,000 to 3,500,000.
5. Mileage-price correlation is 0.013 and year-price correlation is -0.002; neither shows a clear linear pattern.

5. Python code walkthrough

This section follows the supplied Python file in order. The report charts were prepared from the same CSV; the Python file itself was left unchanged.

Code section	What it does	How to read it
Imports and source	Imports pandas, matplotlib and seaborn; loads the CSV from a local Windows path and keeps the Kaggle URL in comments.	The CSV must exist at the coded path for the script to run on this computer.
Shape, fields and types	Prints dimensions, column names and dtypes; records starting rows and separates categorical fields from numeric fields.	Car_ID is an identifier and should not be treated as a vehicle measurement.
Sample and summaries	Displays the first and last five rows, calls info(), and describes numeric columns.	describe() does not summarize categories. print(df.info()) also prints None because info() prints its result and returns None.
Missing values and duplicates	Uses isnull().sum() and duplicated().sum() to check completeness and exact duplicate rows; fill/drop examples are comments.	Comments do not clean data. The checks return zero for this file, so no fill or removal is needed.
Outlier plots	Creates a numeric box plot and a Mileage-by-Brand box plot.	The numeric plot includes Car_ID. The report screens outliers on the six measurements instead.
Brand/fuel counts	Uses value_counts() to find the most and least common brand and fuel type.	The fuel variables are calculated, but the two fuel print statements use the brand variables by mistake.
Price and correlations	Prints Price.describe() and the Mileage/Price correlation table.	Correlation is an association; it does not show cause.
Original charts	Attempts a Brand histogram, Brand-versus-Price box plot, numeric heatmap, and Price/Brand bar plot with Mileage as hue.	Brand is categorical; the heatmap includes Car_ID; many mileage hue groups can be hard to read.
Categories and encoding	Prints unique category labels and uses get_dummies() for Brand, Model, Fuel_Type and Transmission.	Encoding is not needed for this EDA because no model is required. The final info() describes the transformed data.

Conclusion

The file is complete for the checked fields, with 10,000 rows, no missing values, and no exact duplicate records. It also has a model/fuel consistency concern and very weak linear relationships between price and the measured numeric features. It is useful for practicing EDA, while undocumented units and pairings limit claims about the wider car market.

The assignment asks for one code file and one PDF report. This document is the report; the supplied Python file was not changed.